

Hossam Khalil

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Education

PhD in Astrophysics, University of Helsinki (UH), Finland Jan 2025 -Present

Working Title: *Improving cosmological constraints from large-scale structure in the era of eROSITA and Euclid*

Supervisor: Alexis Finoguenov

M.Sc. in Astrophysics, University of Helsinki (UH), Finland Sep 2022–Jun 2024

Title: *AXES-2MRS: A new all-sky catalogue of extended X-ray galaxy groups*

Supervisor: Alexis Finoguenov

B.Sc. in Physics, University of Science & Technology at Zewail City (UST), Egypt Sep 2015–Jul 2019

Grants & Fellowships

Astrophysics Centre for Multimessenger studies in Europe (ACME) Travel Grant

Research visit to the University of Geneva

Sep 2025

University of Helsinki Travel Grant

Research visit to the University of Geneva

Jun 2025

The Vilho, Yrjö and Kalle Väisälä Foundation of the Finnish Academy of Science and Letters Grant

Six-month working grant (EUR 13,500). Declined in favour of a PhD employment contract

Jan 2025

University of Helsinki Scholarship Full M.Sc.tuition-fee waiver (EUR 30,000)

2022–2024

Publications

- **Khalil, H.**, Finoguenov, A., Eckert, D., Seppi, R., Tempel, E., Lovisari, L., and Gastaldello, F. 2026. *eROSITA cosmology with galaxy groups: hot gas budget out to the virial radius*. A&A, accepted [[arXiv](#)]
- **Khalil, H.**, Finoguenov, A., Tempel, E., and Mamon, G. A. 2025. *AXES-2MRS: A new all-sky catalogue of extended X-ray galaxy groups*. A&A, 690, A212 [[arXiv](#)]
- Finoguenov, A., Seppi, R., Eckert, D., **Khalil, H.**, et al. 2026. *Large eROSITA X-ray sources as 2MRS galaxy groups*. A&A accepted [[arXiv](#)]
- Seppi, R., Eckert, D., **Khalil, H.**, et al. 2026. *Bound or blown: the fate of hot gas in galaxy groups*. A&A submitted [[arXiv](#)]
- Eckert, D., Seppi, R., **Khalil, H.**, et al. 2025. *The impact of strong feedback on galaxy group scaling relations*. A&A, 701, A127 [[arXiv](#)]

Teaching

Teaching assistant – Department of Physics, University of Helsinki

2025–Present

- **PAPU-105** Project leading in observational astronomy
- **FYS2045** Basics of Observational Astronomy II

Research Experience

Master Thesis

Department of Physics at University of Helsinki (Supervisor: Alexis Finoguenov)

Jun 2024

Helsinki, Finland

- Released a new all-sky catalogue of X-ray detected groups (AXES-2MRS) by identifying extended X-ray sources in the ROSAT all-sky survey with the 2MRS optical group catalogue. Performed X-ray analysis on a subset observed by XMM-Newton to study the baryonic properties of these groups through X-ray–optical scaling relations.

Research Trainee

Helsinki Institute of Physics & Department of Physics at UH (Supervisor: David J. Weir)

Sep 2023 - Jan 2024

Helsinki, Finland

- Analyzed the NANOGrav 15-year data set to constrain the parameters of a stochastic gravitational wave background signal from a first-order phase transition using PTArcade.

Nordic Optical Telescope (NOT) Remote Observing School

University of Helsinki and University of Turku

Oct 2022

Turku, Finland

- Conducted optical observations of nearby galaxy groups using the NOT in La Palma.
- Carried out the observation preparation, spectral reduction, and redshift analysis to complete cluster surveys.

Zwicky Transient Facility (ZTF) Summer School

University of Minnesota and Caltech

Jul 2022

online

- Performed multi-messenger astrophysics analysis tasks including transient detection in ZTF datasets and optical follow-ups to neutrino (IceCube) and gravitational wave (LIGO & VIRGO) alerts.

Research Trainee in High Energy Astrophysics

UST at Zewail City (Supervisor: Alaa Ibrahim)

Aug 2017 - Jan 2018

Giza, Egypt

- Studied different types of bursts from a radio-emitting magnetar and performed spectral fitting to model the emission region properties.

Professional Experience

Physics Teaching Assistant

UST at Zewail City

Sep 2020 - Jun 2022

Giza, Egypt

- Supervised 50+ students in PEU 327 (Observational Astrophysics Lab) and PEU 450 (Computational Physics).
- Delivered online and in-person lectures, graded assignments, and provided one-on-one support clinics.

Data Analyst

EV Mobility (Mansour Group)

Jan 2020 - Jan 2021

Cairo, Egypt

- Worked in the R&D department, contributing to a project that aimed at introducing electric micromobility solutions to Egypt.
- Analyzed the global electric micromobility scene, conducted market analysis, and provided recommendations that had high-level business consequences.

Scientific Language Editing Trainee

Natural Sciences Publishing

Aug 2019 - Jan 2020

Giza, Egypt

- Worked as a trainee in the Applied Mathematics & Information Sciences Journal and Journal of Statistics Applications & Probability.

Skills

Astrophysics Software

SAS, eSASS, HEASoft, PTArcade, Xspec, IRAF, CARTA, TOPCAT

General Programming Languages

Python, Mathematica, MATLAB

Other Software

MS Office, Adobe Photoshop

Languages

Arabic (Native speaker), English (Fluent), Finnish (Beginner)

REFERENCES ARE AVAILABLE UPON REQUEST